

Chapter 4. Lagrange's Theorem

4.3 Lagrange's Theorem



In this video, we will prove Lagrange's Theorem.

Proving Lagrange's Theorem

THEOREM 4.4. LAGRANGE'S THEOREM. *Let H be a subgroup of a finite group G . Then the order of G is a multiple of the order of H .*

8. There exists a bijective map from H to Ha . In particular, if H is finite, then any coset Ha has the same number of elements as H and therefore all the cosets have the same number of elements.
9. The family of all the cosets of Ha , as a ranges over G , is a partition of G . It means that $G = \cup_{a \in G} Ha$, and $Ha \neq Hb$ implies $Ha \cap Hb = \emptyset$.

Proof. Suppose that $Ha_1, Ha_2, \dots, Ha_k$ are all the distinct cosets of H in G . Then $G \stackrel{(9)}{=} Ha_1 \cup Ha_2 \cup \dots \cup Ha_k$ and $|G| \stackrel{(9)}{=} |Ha_1| + \dots + |Ha_k| \stackrel{(8)}{=} |H| + \dots + |H|$
 $\quad \quad \quad = k \cdot |H|$

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COROLLARY 4.5. The order of any element of a finite group divides the order of the group.

Proof. Suppose that G is a finite group and $g \in G$. By the definition the order of g is the order of the cyclic subgroup generated by g . And by Lagrange's Theorem

$$o(g) = |\langle g \rangle| \mid |G|, \text{ as required} \quad \blacksquare$$

COROLLARY 4.6. Every group of prime order is cyclic.

Proof. Suppose that G is a finite group of order a prime p . Take $g \in G$, ($g \neq 1$) and consider the cyclic subgroup $\langle g \rangle$ generated by g . By Lagrange's Theorem

$$|\langle g \rangle| \mid |G| = p \quad \begin{matrix} \Rightarrow \\ \uparrow \\ |\langle g \rangle| \neq 1 \end{matrix} \quad \langle g \rangle = G, \text{ } G \text{ is cyclic} \quad \blacksquare$$

COROLLARY 4.7. If $|G| = p^2$ where p is a prime, then either G is cyclic or $g^p = 1$ for every element g in G .

Proof. Suppose that G is not cyclic, and take $g \in G, g \neq 1$. We want to prove that $g^p = 1$. Consider the cyclic subgroup $\langle g \rangle$ generated by g . By Corollary 4.5 we have that

$$o(g) \mid |G| = p^2 \Rightarrow o(g) \in \{1, p, p^2\}.$$

Notice that $o(g) \neq 1$ because $g \neq 1$; and $o(g) \neq p^2$ because $o(g) = p^2$ would imply that $\langle g \rangle = G$ and we are assuming G not cyclic. So, $o(g) = p$ and then $g^p = 1$ $\square$

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Definition 4.8. Let G be a group and H a subgroup of G . The **index of H in G** is the number of cosets of H in G . We denote it by $|G : H|$.

Remark 4.9. If G is finite, from the proof of Lagrange's Theorem, we get that the number of elements in G is equal to the number of elements in H , multiplied by the number of cosets of H in G . In other words:

$$|G : H| = \frac{\text{order of } G}{\text{order of } H} = \frac{|G|}{|H|}$$

A subgroup H can have finite index in a group G even if both H and G are infinite. For example, $H = 4\mathbb{Z}$ has index 4 in $G = \mathbb{Z}$ and both $4\mathbb{Z}$ and $\mathbb{Z}$ are infinite.

$$|G| = \underbrace{\left(\begin{array}{c} \text{number of} \\ \text{cosets of } H \text{ in } G \end{array} \right)}_{\text{index}} \cdot |H|$$

$$|G| = |G : H| \cdot |H|$$